

Solution Requirements

Date	7 July 2026
Team ID	XXXXXX
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can securely register and log in using email and password with JWT-based authentication.</i>	High
2	Authorization levels	<i>Only authenticated users can access financial information, loan management, dashboard, and AI modules.</i>	High

3	External interfaces	<i>The system communicates with the Google Gemini API</i>	High
		<i>(or rule-based AI), SQLite database, and REST APIs using Axios.</i>	High
4	Transactions processing	<i>The system processes user registration, login, loan CRUD operations, financial calculations, settlement prediction, and AI-generated negotiation emails.</i>	High
5	Reporting	<i>The dashboard displays monthly income, expenses, debt ratio, financial health score, settlement prediction, and history reports.</i>	Medium
6	Business rules, etc.	<i>Monthly surplus is calculated as $\text{Income} - \text{Expenses}$. Debt ratio and settlement prediction are generated from the user's financial profile and loan details.</i>	High
7	Compliance to laws or regulations	<i>The system follows secure authentication practices, password hashing, JWT security, and borrower rights information based on RBI guidelines.</i>	Medium

8	Other	The application provides AI-assisted financial guidance, negotiation email generation, and borrower rights information through an intuitive user interface.	Medium
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Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	Backend APIs and dashboard should respond quickly for smooth user interaction.	Page/API response within 2 seconds
2	Scalability	The system should support increasing numbers of users and financial records without major redesign.	Supports 500+ concurrent users
3	Security & Data Privacy	Protect user information using password hashing, JWT authentication, secure API communication, and input validation.	Secure authentication and encrypted password storage
4	Reliability & Availability	The application should remain available with stable backend services and proper exception handling.	99% uptime during normal operation
5	Usability & Accessibility	The interface should be responsive, user-friendly, and compatible with desktop and mobile devices.	Responsive design with easy navigation
S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

6	Other	<i>The system should be modular, maintainable, and easy to extend with future AI features, cloud deployment, and database upgrades.</i>	Clean modular architecture with reusable components
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